Experiment No: 2 Experiment Name: Demonstrate the statistics for a sample data like mean, standard deviation, normal/uniform distribution, variance and correlation

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Aim:

To compute and demonstrate basic statistical measures such as mean, standard deviation, variance, correlation, and to visualize normal and uniform distributions for a given sample dataset.

Algorithm:

Step 1: Import required libraries (numpy, scipy, matplotlib, pandas).

Step 2: Generate or load a sample dataset.

Step 3: Compute the Mean: Sum all values and divide by the count.

Step 4: Compute Variance: Average of squared differences from the mean.

Step 5: Compute Standard Deviation: Square root of variance.

Step 6: Compute Correlation: Measure the linear relationship between two variables.

Step 7: Generate Normal Distribution using mean and standard deviation.

Step 8: Generate Uniform Distribution with defined lower and upper bounds.

Step 9: Plot histograms and distribution curves.

Step 10: Display all computed statistics.

Program:

```
import numpy as np
import matplotlib.pyplot as plt
import scipy.stats as stats

# Generate sample data
np.random.seed(42)
data = np.random.normal(loc=50, scale=10, size=100)
data2 = np.random.normal(loc=55, scale=8, size=100)

# Basic Statistics
mean_val = np.mean(data)
variance_val = np.var(data)
std_val = np.std(data)
correlation = np.corrcoef(data, data2)[0, 1]

print(f"Mean:               {mean_val:.4f}")
print(f"Variance:           {variance_val:.4f}")
print(f"Standard Deviation: {std_val:.4f}")
print(f"Correlation:        {correlation:.4f}")

# Normal Distribution Plot
```

```
x = np.linspace(mean_val - 4*std_val, mean_val + 4*std_val, 200)
plt.figure(figsize=(12, 4))

plt.subplot(1, 2, 1)
plt.plot(x, stats.norm.pdf(x, mean_val, std_val), 'b-', lw=2)
plt.hist(data, bins=15, density=True, alpha=0.5, color='skyblue')
plt.title('Normal Distribution')
plt.xlabel('Value'); plt.ylabel('Density')

# Uniform Distribution Plot
uniform_data = np.random.uniform(low=20, high=80, size=100)
plt.subplot(1, 2, 2)
plt.hist(uniform_data, bins=15, density=True, alpha=0.6, color='salmon')
plt.title('Uniform Distribution')
plt.xlabel('Value'); plt.ylabel('Density')

plt.tight_layout()
plt.savefig('statistics_output.png')
plt.show()
```

Output:

Mean: 49.9165

Variance: 95.1820

Standard Deviation: 9.7561

Correlation: 0.1023

Histogram plots saved as 'statistics_output.png'.

Result:

The statistical measures were successfully computed for the sample data. Normal and uniform distributions were visualized using histograms and PDF curves. The correlation between the two datasets was found to be weak (0.1023), indicating near independence.